

Sibyl: Explaining Machine Learning Models for High-Stakes Decision Making

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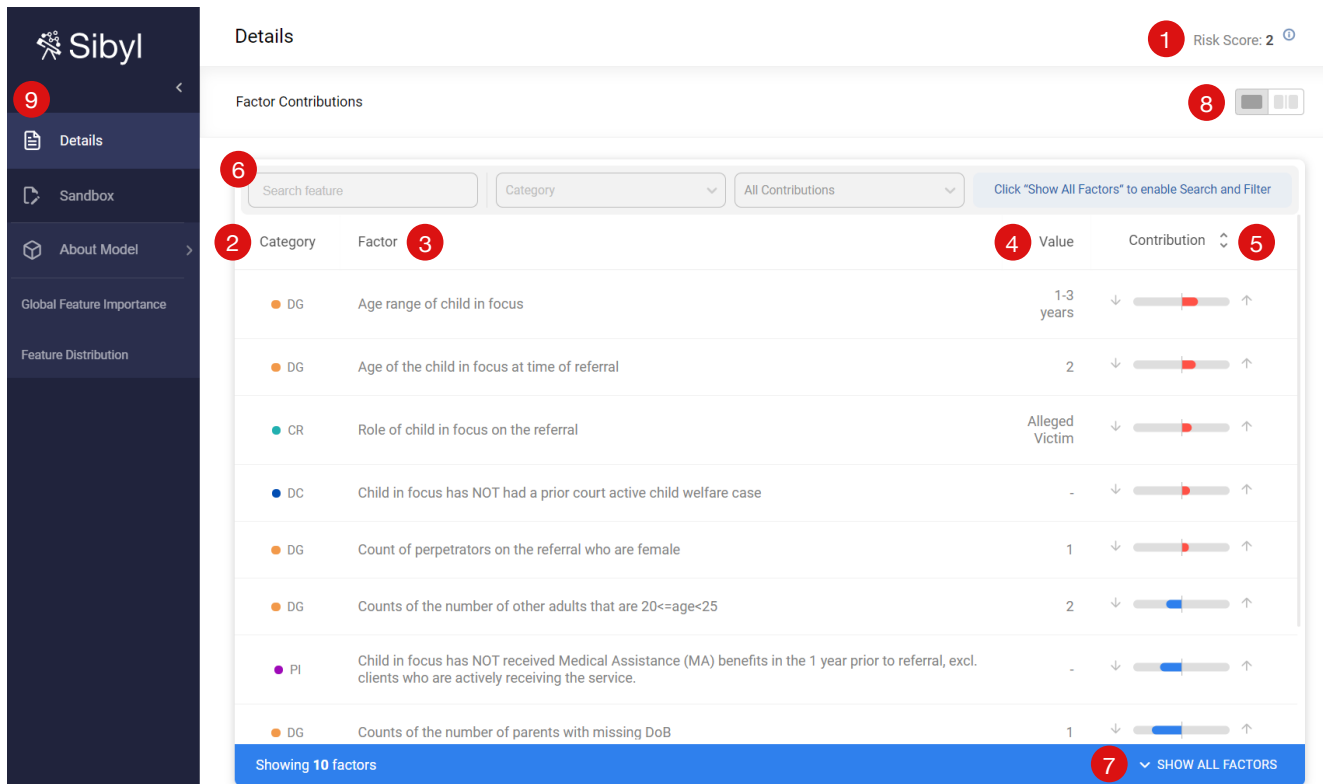


Figure 1: The SIBYL case-specific details interface. This interface shows feature contributions for child welfare ML model predictions. Note that in the interface, we refer to features as “factors”, as this language is more familiar to our users. 1) The risk score for the case (1-20). 2) Categories for each feature, such as demographics (DG) or referral history. 3) A short description of each feature. 4) The value of numeric or categorical features. 5) The contribution of each feature (the table can be sorted in ascending or descending order of contribution). 6) UI components for searching by feature name or filtering by category, enabled when the full feature list is shown. 7) A button to switch between showing only the top 10 most contributing features to showing all features. 8) A button to switch between single-table view and a side-by-side view, which splits features that increase and decrease risk. 9) A sidebar that includes other explanation types, as described in this paper.



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CHI '21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan
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ACM ISBN 978-1-4503-8095-9/21/05.
<https://doi.org/10.1145/3411763.3451743>

ABSTRACT

As machine learning is applied to an increasingly large number of domains, the need for an effective way to explain its predictions grows apace. In the domain of child welfare screening, machine learning offers a promising method of consolidating the large amount of data that screeners must look at, potentially improving the outcomes for children reported to child welfare departments.

Interviews and case-studies suggest that adding an explanation alongside the model prediction may result in better outcomes, but it is not obvious what kind of explanation would be most useful in this context. Through a series of interviews and user studies, we developed SIBYL, a machine learning explanation dashboard specifically designed to aid child welfare screeners' decision making. When testing SIBYL, we evaluated four different explanation types, and based on this evaluation, decided a local feature contribution approach was most useful to screeners.

CCS CONCEPTS

• **Human-centered computing** → **Field studies**; *User studies*.

KEYWORDS

machine learning, interpretability, explainability, child welfare, social good, tool

ACM Reference Format:

Alexandra Zytek, Dongyu Liu, Rhema Vaithianathan, and Kalyan Veeramachaneni. 2021. Sibyl: Explaining Machine Learning Models for High-Stakes Decision Making. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3411763.3451743>

1 INTRODUCTION

Thanks to innovations in machine learning (ML), computers can now help with many tasks previously performed by humans alone, often improving both speed and performance. However, in many domains, human decision-makers provide essential insights that cannot be replaced by existing algorithms. In such cases, decision-making is improved when ML output is used to augment human decision making.

ML models often output only a single number or classification, such as the risk score seen in the upper right corner of figure 1. This can make it difficult for human decision-makers to trust the model and handle disagreements between their opinions and the model output. Through the use of ML augmentation interfaces, human decision-makers can continue to think critically about the task, while also taking advantage of the added information provided by the model.

The field of ML explainability has attempted to foster this human-computer interaction and offer such augmentation by offering explicit explanations for a model's decisions. However, much existing work on ML explainability is aimed at an audience of ML/data experts [14] [10] [5] or domain experts in technical or data-driven fields such as medicine [9] [7]. For example, Zhang et. al. [14] developed an explanation framework for helping data scientists and ML experts interpret and debug ML models, and Lundberg et. al. [9] developed an explanation interface for helping anaesthesiologists prevent hypoxaemia during surgery. Many fields, however, are more qualitative in nature, with decisions following discussion more than data crunching.

In this paper, we investigate the usage of an ML model for decision-making in one such qualitative field: the screening of maltreatment referrals made to child welfare services. These are high-stakes decisions: failure to investigate actual child abuse can

be fatal, while imposing the burden of investigation on a family where child abuse is not present is needlessly stressful and invasive. Recently, child welfare systems in some jurisdictions have been looking at machine learning tools to aid in these difficult decisions [3]. However, interviews conducted with screeners — described in this paper — have suggested that these tools may be more helpful with additional explanations

To better facilitate and experiment with providing explanations in this field, we have developed SIBYL, a machine learning explanation dashboard designed to allow humans to work with algorithms to make decisions. SIBYL offers several explanation interfaces, selected to be easy to parse even by users with no technical expertise, while still providing an accurate representation of the model.

Before and throughout the development of SIBYL, we held multiple interviews with child welfare call screeners, and observed the existing decision-making process. After implementing SIBYL, we performed a user study. All of these steps revealed what the child welfare screeners really need from explanations, and what is still missing from the literature.

1.1 Contributions

Our work has the following contributions:

- (1) We followed an iterative, collaborative procedure to provide explanations of machine learning predictions in the domain of child welfare screening. In doing so, we worked towards the end-goal of deploying it in the child welfare domain.
- (2) We began the process of analyzing and addressing the explanation needs and challenges that arise when trying to effectively explain ML predictions to child welfare experts, and similar non-technical domain experts making decisions in high-risk fields.

2 UNDERSTANDING CONTEXT AND END-USER NEEDS

Before beginning work on SIBYL, we took several steps to understand the use-case and user needs. In this section, we discuss some relevant background on the context, as well as our goals and findings from these first steps.

2.1 Child Welfare Screening

Child abuse is an active issue in communities. The Center for Disease Control and Prevention estimates at least 1 in 7 children have experienced child abuse and/or neglect in the past year [1]. More than one-third of American children are investigated for abuse or neglect by age 18 [6]. Still, in 2018, there were 1,770 reported fatalities resulting from child abuse and neglect [2].

In the U.S., regional Child Protective Services (CPS) agencies are tasked with handling child abuse and neglect referrals from concerned members of the community, as well as mandated reporters such as teachers, who are required by law to report any suspicion of abuse or neglect. These referrals are examined by child welfare specialists ("call screeners"), who decide whether to screen in or screen out each case. A screened-in case will be investigated further, while a screened-out case will be recorded but not investigated.

In recent years, predictive risk modelling (PRM) has been deployed in child welfare contexts in multiple counties, with the goal

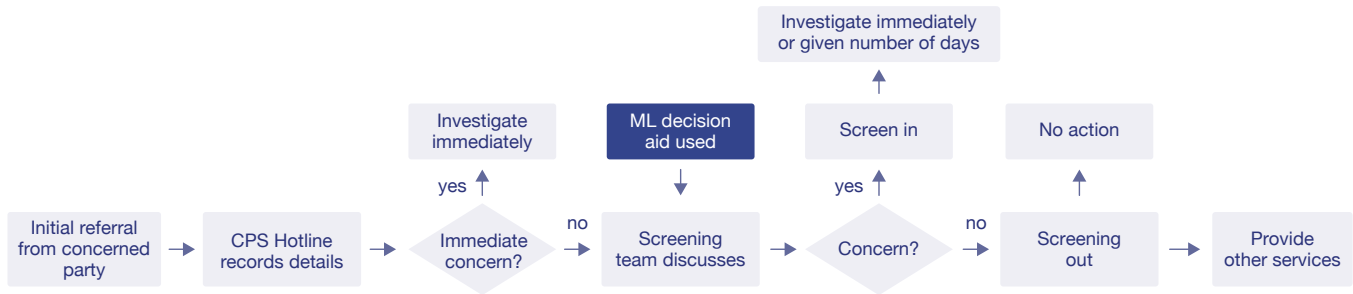


Figure 2: The general child welfare screening process used by the CPS department of our collaborating county.

of enabling more efficient and consistent decision-making and improving the overall health and safety of county residents [12]. One example of such a model was deployed in Allegheny County, PA by Vaithianathan et. al. [12]

Currently, PRM is being introduced to Larimer County, CO by Vaithianathan et. al. [11], through a LASSO linear regression model trained on 461 features, which include information such as the child and parents’ age, past referrals and their outcome, and past court involvements. The model predicts the likelihood of removal from home in the next two years [12]. This paper focuses on the usage of this model.

2.2 Study Participants

We collaborated over the course of a year with a pool of 19 social workers and supervisors working for the child welfare department in a collaborating county in Colorado. All participants regularly act as screeners in the county’s child welfare screening decision making process.

Our collaborations began in December 2019, with two days of in-person field observations (section 2.3). Following this, we observed a simulated case review session via video conferencing (section 2.3), and conducted several interviews also via video conferencing (sections 2.4 and 3.1). Finally, our collaborating screeners participated in our user study digitally (section 4).

2.3 Observing the task

To better understand the existing child welfare screening workflow, we travelled to our collaborating county in Colorado to observe screeners’ decision-making on referrals without using the ML model. This process revealed:

- (1) Our collaborating county uses the general procedure for child welfare screening shown in Figure 2. In cases of immediate concern, a referral may be screened in immediately after it is received by CPS (this decision is made by a child welfare supervisor). In most cases, however, the decision as to whether to screen-in or screen-out a referral is made by a team of child welfare experts. It is this team that receives the ML risk score prediction.
- (2) Five to ten minutes are spent on each case by this team. Most of this time is spent going over the details of the case. The screening decision is made after about one to two minutes of discussion.

- (3) A large portion of this time is dedicated to determining the factors that are associated with higher and lower likelihood of abuse — referred to as risk and protective factors, respectively — involved in a case, and weighing these against each other.

To identify the explanation needs present in the domain of child welfare screening, we observed a simulated case review session using the ML model, where our collaborating screeners (split into 3 teams, labelled **T1**, **T2**, and **T3**) were asked to make decisions about historical referrals as they normally would — but with the addition of ML model scores, which were provided before they made the final screening decision. Each team made decisions on 7 to 9 cases. The participants were then interviewed and asked to reflect upon how they used each score, whether the scores aligned with their expectations, and how the scores impacted their decisions.

Our findings from this step included:

- (1) Screeners expressed a lack of trust when making decisions using the ML model alone, evidenced by their tendency to not consider the model prediction at all when it disagreed with their intuition. For example, when asked if the score caused them to reconsider their decision, **T3** responded “No, [we were] surprised it is that low.”
- (2) Screeners did not have a clear path forward when they disagreed with the model prediction, sometimes electing to ignore it entirely (see previous item) and sometimes trying to justify it based on how they thought the model worked. For example, **T2** reported in one case that the score made them “think a little deeper about why the score is so high [and caused us to] take another look at [the history]”
- (3) Because the model provides auxiliary information (1-20 risk score based on the likelihood of removal from home in 2 years) rather than a direct decision suggestion, there was some confusion about how to use the model prediction target. For example, when asked how the model affected their decision making process, **T3** responded “[we did not know] enough of what the score means to know how to accurately use it.”
- (4) As expected for such a sensitive domain, users were concerned about the ethics of using the ML model score. There was concern that the model may prevent critical thinking. **T3** commented

“[The model] could be dangerous for people just looking at the number, need to take everything into account. Makes you stop and think and ask yourself are you critically thinking...”

2.4 Interviewing and proposing explanations

To get concrete domain-expert insight, we interviewed screeners about what kind of additional information they would be interested in alongside ML predictions, and proposed possible options.

Our findings from these steps included:

- (1) Screeners were confident that they would want to know why the model made the predictions it made.
- (2) Screeners believed that understanding how important each feature was to the score prediction would be helpful.
- (3) One screener requested a description of what steps the model takes in making predictions.
- (4) Screeners were interested in getting “what-if” style explanations that gave information about what could be changed about a child to reduce his or her risk. Note that the ML model is not trained on causal relationships, so explanations would not be able to provide such information.
- (5) Some screeners were interested in seeing similar cases they dealt with in the past. Others thought this would be too much information to digest in such a short period of time.

3 DESIGNING SIBYL

Having established our user needs, we shifted our focus to designing a first draft of SIBYL. To collect early feedback on our design from end-users, we created mockups of explanation interfaces that could be added to SIBYL.

In this section, we describe the complete, original SIBYL design. The design includes four different explanation interfaces, each with a separate purpose and goal.

Case-Specific Details: Local Feature Contributions The *Case-Specific Details* interface provides a simple, accurate interpretation of where an individual model prediction comes from through feature contributions. The table in this interface assigns each feature a contribution, either positive or negative. This interface can be seen in figure 1.

The local feature contributions were found using the Shapley Additive Explanations (SHAP) algorithm [8]. We chose to use SHAP because it is a theoretically sound approach that generates consistent and intuitive explanations for a machine learning model.

A possible source of confusion stems from the phrases *positive contribution* and *negative contribution*. In the machine learning community, a positive contribution indicates that the model prediction will increase in value. In the screener community, however, an increased risk score is only a negative occurrence. Following discussion during the design phase, the terms “positive” and “negative” are never used in the app or instructions.

Sandbox: Investigating “What Ifs” The *Sandbox* interface allows users to experiment with and see how the model prediction would change if features differed. It has two parts.

The *Experiment with Changes* box allows users to change up to four feature values at a time, to investigate specific “what-if” questions they may have. The *Model predictions if each value was*

changed box shows the resulting prediction if each Boolean feature value was individually reversed.

About Model: Global Feature Importances The *About Model* interfaces offer explanations of the model’s general logic, outside of the context of a individual prediction. These interfaces were added with the intention of allowing users to calibrate their trust in the model and learn about its strengths and weaknesses.

The first *About Model* interface is the *Global Feature Importance* explanation. This interface shows the general, relative importance of each feature. It also provides a brief description of the model architecture and logic, as well as its performance metrics.

The global feature importances were found using the Permutation Importance algorithm [4]. This algorithm computes the change in model performance if each feature is permuted individually. It therefore describes how closely each feature is linked to model performance.

About Model: Feature Distributions The second *About Model* interface is the *Feature Distributions* explanation. This interface shows the distribution of feature values among past cases of a given score. This explanation gives a quick retrospective view of how the model performed in the past. It also provides the number of cases that resulted in a removal from home for each risk score.

3.1 Gathering and Iterating on Mockup Feedback

Next, we showed the high-fidelity mock-ups of the five explanation interfaces to a group of about 20 screeners, who provided feedback. Our goal from this step was to get additional, domain-expert insights on the proposed explanations, giving them a final chance to give concrete, grounded feedback on the designs before implementation.

Our findings from this step included:

- (1) Screeners said that in their usual workflow, they would list “Risk” and “Protective” factors side-by-side. To mirror this, the updated **Details** interface has a split-view toggle that shows the negative- and positive-contribution features in two side-by-side tables.
- (2) Feedback received confirmed that the **Sandbox** interface has a risk of being misconstrued as suggesting action or reflecting real-world causal structures.
- (3) It was suggested that the **About Model** interfaces may be more useful for training or education, and for building screener trust in the model before using it, rather than provided during active decision making.

4 USER STUDY

To evaluate SIBYL, we ran two user studies — first with non-expert participants (people without experience in child welfare screening) and then with expert participants (child welfare screeners). For brevity, we discuss the results of both user studies together in this section. Small usability fixes were made to the SIBYL design after the first user study based on feedback, and otherwise the studies had the same format.

In total, we had 13 expert participants and 12 non-expert participants. The expert participants were social workers with experience in child welfare screening, and the non-expert participants included

Table 1: Summary of answers given by expert participants to the question “what caused you to trust the model more or less”. The top table lists reasons for having “a great deal”, “a lot” or “a moderate amount” of trust in the model. The bottom half list reasons for having “a little”, “not at all” trust in the model.

Factors that increased trust		
Category	# of answers	Sample answers
General comments about being shown protective and risk factors (Details page)	8	"The details and the risk and protective factors and the contribution they have", "Info in the details and risk factors"
Specific factors listed as risk or protective (Details page)	8	"The past number of child welfare involvements (listed in the features listed)", "The risk factors involved, especially prior placements, benefits, and current CPS involvement"
The number of factors listed	5	"Very few risk factors", "The lack of protective factors"
The score agreed with screener intuition	10	"... the children are not residing with the alleged perpetrators which I would assume would reduce the risk score", "Model prediction makes sense"
The explanation agreed with screener intuition	3	"Risk factors made sense for the model prediction number"
General comments about sandbox page	1	"Details under sandbox of why the risk level was so high"
General comments about the explanation providing more information or understanding	6	"Allowed for more understanding", "History clarification"
Factors that decreased trust		
Category	# of answers	Sample answers
A specific, key factor was not considered by the model	3	"This is a case for law enforcement, not CPS", "... it may have been handled during the open case"
The importance weighting of factors was off	2	"Young, vulnerable children being left alone is still cause for concern, despite past involvement"
The score disagreed with screener intuition	3	"Seems high, no [health history], no real proof of any drug use, no proof child is at any risk of abuse"
There was some confusion about information presented	2	"There were discrepancies between the info in the referral and the info provided by the tool"
The screener wanted more information	2	"I would want to see other referrals for the family"

data and social scientists. 2 of the expert participants completed the task while video conferencing and screen-sharing with experimenters.

For data privacy reasons, all data used in this section was synthetic or deidentified. Due to the nature of the problem, we could not effectively give a ground truth value to synthetic data, so we could not quantify change in outcome. Therefore, the main focus for this study was collecting qualitative feedback on the usability and usefulness of the explanations, as well as determining which explanations would be the most useful.

4.1 Study Procedure

Participants were first shown a short video explaining how to use SIBYL. Next, they were shown 7 de-identified case descriptions, accompanied by the model’s prediction and SIBYL explanation. Not all participants completed all 7 cases. In total, experts completed 73 individual case analyses, and non-experts completed 75.

For each case, users were asked to make a screen-in or screen-out decision. Then, they answered a series of reflection questions about their experience using the explanation tool and trust in the model.

4.2 Study Results

4.2.1 Tool Usability. An important goal of our user study was to judge the usability of the explanation methods, and make improvements for future iterations of SIBYL.

Most usability concerns resulted from our explanations being too transparent to the model — in other words, presenting information

about features in the form that the model uses, rather than the form most familiar to humans.

Our updated version of SIBYL uses language more similar to what the screeners use. This includes 1) replacing Boolean True / False terminology with positive and negative sentences, 2) combining correlated factors into single concepts, and 3) greatly reducing the number of factors shown by default (10 most contributing factors versus the original 400+).

4.2.2 Helpful Interfaces. The Details interface was by a large margin considered the most helpful, by both experts and non-experts. It was labelled as being helpful by experts in 91.8% of case analyses, and by non-experts in 90.7% of case analyses. This was significantly higher than Sandbox (experts: 16.4%, non-experts: 22.6%) and Feature Distributions (experts: 20.5%, non-experts: 8.0%). Feature Importance was never listed as helpful by either group.

4.2.3 Model Trust. The explanations were reported to both increase and decrease perceived user trust, for different reasons. Table 1 lists the common reasons given by expert participants for trusting the model more or less, respectively. We see that agreeing with the model’s score improves perceived trust the most. Beyond this, the details interface was frequently cited as improving perceived trust, either due to specific factors listed or more general elements of the interface, such as the number of factors. Trust was reduced when there was confusion or inconsistencies in the presented information, or when the model did not consider important factors that participants knew about.

5 DISCUSSION

5.1 Key Findings

Here, we summarize our findings so far in our work in applying ML explanations to the domain of child welfare.

The need that appeared to be most significant for the child welfare experts was to learn more, pertinent information about the *case at hand*. They wanted explanations to be true to reality. We therefore suggest that in similar contexts, *accuracy about reality* is more important than *fidelity about the model*. If it is impossible to have explanations with a reasonable degree of such accuracy, the model may not be appropriate for the use-case.

Similarly, the users expressed almost no interest in learning about the model itself. Only once (see section 2.4, item 3) did any domain expert ever express interest in understanding the details of how the model worked under the hood — and even then, they were mostly looking for a sanity-check style overview. Much of the feedback we received on the first version of our user toolkit was related to our explanation being *too transparent*, ie. providing all features in their model-ready forms.

Finally, we observed that explanations are capable of encouraging cognitive biases, just as they may mitigate them [13]. For example, our sandbox interface appeared to encourage interpreting correlation as causation.

5.2 Future Work

Our next step will be to review our findings to determine what generalizable lessons can be found for other, non-technical/high-risk domains.

We will also work to implement SIBYL in the real world, and improving decision-making in live deployment. We will continue to empirically validate any changes in outcome resulting from the use of SIBYL through user studies that ask screeners to make decisions on real, historic cases. Once this validation is complete, and after making any necessary design iterations, we will begin A/B testing on live decision-making.

ACKNOWLEDGMENTS

We would like to thank Diana Benavides Prado, Megh Mayur, Athena Ning, and Larissa Lorimer for their guidance throughout the process of applying machine learning explainability to child welfare, for connecting us to the child welfare domain experts, and for providing us with their trained model and dataset. We thank the child welfare experts at Larimer County Department of Human Services for their invaluable insights and feedback on the tool. We thank Iulia Ionescu, Sergiu Ojoc, Ionut Radu, and Ionut Margarint for their work on developing the Sibyl application. We thank Arash Akhgari for his work on our diagrams and visualizations. We thank

Michaela Henry for support in project management and insights. We thank the participants of the user studies for their time and feedback. We would like to thank our anonymous reviewers for their comments and suggestions.

This work was supported in part by National Science Foundation grant #1761812 - Spokes: MEDIUM: NORTHEAST: Collaborative Research: Data Science Foundry: A Collaborative Platform for Computational Social Science.

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